Heat Flow in a Thin Rod

A texwright showcase

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October 7, 2026

Abstract

This document exercises what texwright does without a TeX installation: numbered equations with per-row labels, theorems and proofs, figures, tables, cross-references, citations from BibTeX, code, callouts, footnotes and a table of contents with real page numbers. The source is plain Markdown with a little LaTeX mixed in.

Keywords: heat equation, Fourier series, typesetting

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1 Introduction

Fourier introduced the heat equation in 1822 [1] and with it the series that now carry his name. We follow the modern treatment of Evans [2] and compare an exact solution with the Crank–Nicolson scheme [3, 4].

The problem sits on the unit interval. Section 2 states it, Section 3 solves it, and Section 4 compares the result with a numerical scheme.[1]

2 The model

Let $u(x,t)$ be the temperature of a rod of length one with diffusivity $\alpha > 0$. Then

$$u_t = \alpha\, u_{xx}, \qquad 0 < x < 1,\ t > 0, \tag{1}$$

$$u(0,t) = u(1,t) = 0, \qquad t > 0, \tag{2}$$

$$u(x,0) = f(x), \qquad 0 \le x \le 1.$$

The boundary conditions (2) hold the ends at zero. Energy decays:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\frac{1}{2}\,\|u\|_{L^2}^2 = -\alpha\,\|u_x\|_{L^2}^2 \le 0. \tag{3}$$

[1] Fourier's memoir was first submitted to the Institut de France in 1807 and published as a book only fifteen years later.

Theorem 1 (Uniqueness). *Problem (1)–(2) with initial data $f \in L^2(0,1)$ has at most one solution.*

Proof. If u and v solve the problem, $w = u - v$ solves it with $f = 0$. By Equation (3), $\|w(\cdot,t)\|_{L^2} \le \|w(\cdot,0)\|_{L^2} = 0$ for all t. □

3 Exact solution

Separation of variables gives

$$u(x,t) = \sum_{k=1}^{\infty} b_k \, e^{-\alpha (k\pi)^2 t} \sin(k\pi x), \qquad b_k = 2\int_0^1 f(x)\sin(k\pi x)\,\mathrm{d}x. \tag{4}$$

For $f \equiv 1$ only odd modes survive, with $b_k = 4/(k\pi)$. Figure 1 shows how quickly the higher modes die out.

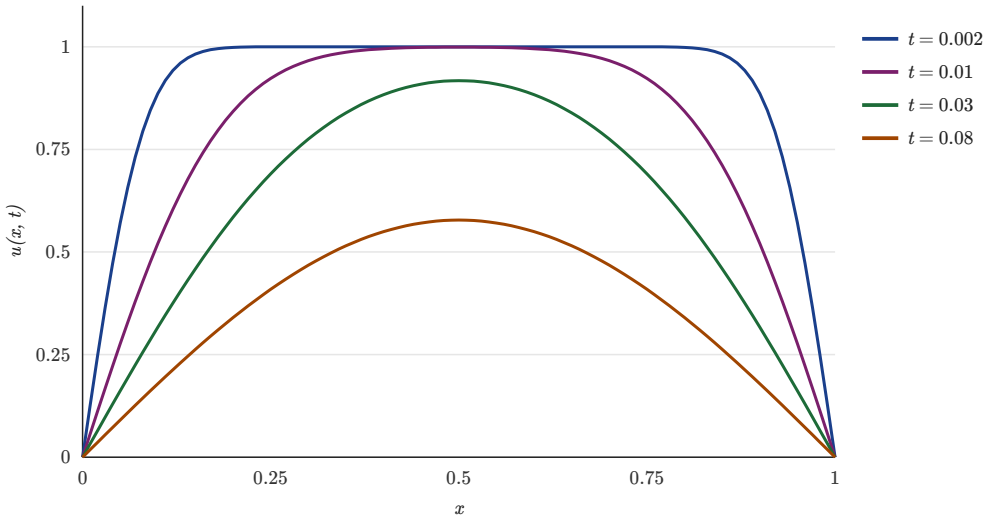


Figure 1: Temperature profiles $u(x,t)$ for $f \equiv 1$ and $\alpha = 1$. High frequencies vanish first, so the profile becomes a single sine arch.

Theorem 2 (Exponential decay). *For $f \in L^2(0,1)$, $\|u(\cdot,t)\|_{L^2} \le e^{-\alpha\pi^2 t}\,\|f\|_{L^2}$.*

> **Tip**
> Theorem 2 follows from (4) and Parseval's identity, since every mode decays at least as fast as the first.

4 Numerical comparison

The Crank–Nicolson scheme is second order in space and time. Table 1 lists the maximum error at $t = 0.1$.

The scheme fits in a few lines:

Listing 1: One Crank–Nicolson step

```
import numpy as np
from scipy.linalg import solve_banded
```

Table 1: Crank–Nicolson errors at $t = 0.1$.

N	Δt	max error	rate
16	10^{-2}	3.1×10^{-3}	—
32	5×10^{-3}	7.8×10^{-4}	1.99
64	2.5×10^{-3}	1.9×10^{-4}	2.00
128	1.25×10^{-3}	4.9×10^{-5}	2.00

```
def step(u, r):
    """Advance u by one step; r = alpha * dt / dx**2."""
    n = len(u)
    ab = np.zeros((3, n))
    ab[0, 1:], ab[1, :], ab[2, :-1] = -r / 2, 1 + r, -r / 2
    rhs = (1 - r) * u
    rhs[1:] += r / 2 * u[:-1]
    rhs[:-1] += r / 2 * u[1:]
    return solve_banded((1, 1), ab, rhs)
```

Listing 1 solves one tridiagonal system per step, so the cost is linear in N.

> **Note**
> Every equation, figure, table and listing number in this document, and every page number in the contents, was computed by texwright. The source contains none of them. Equation (1) is on page 1.

5 Conclusion

Typesetting a document like this used to mean installing TeX. texwright needs neither TeX nor a browser: it breaks paragraphs with the Knuth–Plass algorithm [5], lays out the mathematics by TeX's rules, and writes the PDF itself. When something is wrong, it says which line and why.

References

[1] J.-B. J. Fourier, *Théorie analytique de la chaleur*. Paris: Firmin Didot, 1822.

[2] L. C. Evans, *Partial Differential Equations*, 2nd ed. Providence, RI: American Mathematical Society, 2010.

[3] J. Crank and P. Nicolson, "A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type," *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 43, no. 1, pp. 50–67, 1947, doi: 10.1017/S0305004100023197.

[4] R. J. LeVeque et al., "Finite difference methods for ordinary and partial differential equations," *Journal of Computational Physics*, vol. 227, pp. 1–20, 2007.

[5] D. E. Knuth and M. F. Plass, "Breaking paragraphs into lines," in *Software: Practice and Experience*, 1981, pp. 1119–1184.